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To cite this article: Jane C. Miller (1969) Optical properties of liquid metals at high temperatures, Philosophical Magazine, 20:168, 1115-1132, DOI: [10.1080/14786436908228198](https://doi.org/10.1080/14786436908228198)

To link to this article: <http://dx.doi.org/10.1080/14786436908228198>



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Optical Properties of Liquid Metals at High Temperatures

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[Received 25 July 1969]

ABSTRACT

Measurements have been made, by a polarimetric method, of the optical properties of seven liquid metals, in the frequency range between 0.77 and 5.0 eV and at temperatures up to 1600°C. For Al the results show no sign of the absorption peak at about 1.5 eV which is observed for the solid and are in rough agreement with the Drude free-electron curves; deviations from these curves may be due to surface contamination, though an attempt to explain them quantitatively in this way is not successful. For Cu, Ag and Au the results at low frequencies suggest that the effective density of conduction electrons, n^* , rises slightly on melting, as is to be expected if the structure of the conduction band is smoothed out; the absorption edges due to excitation of electrons from the d band, however, are scarcely changed at all. For the transition metals Fe, Co and Ni, which lie next to one another in the Periodic Table, the absorption spectra show little in the way of structure; the most notable feature of the results is that the magnitude of the absorption is much the same for all three.

§ 1. INTRODUCTION

PREVIOUS measurements of the optical properties of liquid metals have been largely confined to those which melt at easily accessible temperatures, and, with the possible exception of Hg, the results are in reasonable agreement with the predictions of free-electron theory (see Faber 1966, for a review). That is to say, the observed values for the real and imaginary parts of the complex dielectric constant $\epsilon (= \epsilon_1 + i\epsilon_2)$ have been adequately fitted by the Drude formulae

$$\epsilon_1 = 1 - \frac{4\pi ne^2\tau^2}{m(1 + \omega^2\tau^2)}, \quad \epsilon_2 = \frac{4\pi ne^2\tau}{m\omega(1 + \omega^2\tau^2)} \quad \dots \quad (1)$$

with a relaxation time τ independent of frequency ω . To obtain the best fit, not only over the visible range where $\omega\tau$ is usually of order unity but also to the d.c. conductivity (this is the limit of $\omega\epsilon_2/4\pi$ as ω goes to zero) one may need to use an effective carrier density n^* which is rather greater than the density n of valence electrons, but an explanation for this sort of deviation is available, as we shall see below. There is usually no sign of the absorption peaks which are observed for the same metals in the solid and which are associated with band structure above the Fermi level.

This paper describes optical measurements with an apparatus which allowed the specimens to be heated up to 1600°C and the frequency to be varied from the near infra-red to the near ultra-violet. The metals investigated were Al, Cu, Ag, Au, Fe, Co and Ni. A few results have been reported previously for liquid Cu, Ag and Au (references are given in § 4) but they are incomplete, and for the other liquid metals in the list there are no useful data in the literature. Al is a good example of a metal which in the solid state shows a prominent absorption peak, undoubtedly associated in some way with electronic band structure. Cu, Ag and Au are of particular interest because of the excess absorption they display in the ultra-violet due to excitation of electrons from the d band, which is only a few eV below the Fermi level; is the d band shifted or broadened by the melting process? And from the noble metals it is a natural step to the transition metals, in which the d band overlaps the Fermi level.

§ 2. PRINCIPLE OF THE METHOD

The measurements were made by an ellipsometric method adapted from Beattie and Conn (1955). A collimated beam of light was passed through a polarizer, reflected from the specimen surface at a large angle of incidence, passed through an analyser and a monochromator, and then allowed to strike a detector. If the settings of the polarizer and analyser are described by angles P and Q , the output from the detector should be proportional to :

$$\rho^2 \cos^2 P \cos^2 Q + \sin^2 P \sin^2 Q + 2\rho \cos \Delta \sin P \sin Q \cos P \cos Q,$$

where $\rho \exp(i\Delta)$ is the ratio between the amplitude reflection coefficients for the p and s components. Thus when the analyser is set so that Q satisfies the equation :

$$\tan Q = \rho[1 \pm \sqrt{(1 + \cos^2 \Delta)}]$$

the output should be the same for the two settings of the polarizer corresponding to $P=0$ and $\pi/4$. It was a straightforward matter experimentally to identify two different analyser settings which had this property and to determine ψ ($=\tan^{-1} \rho$) and Δ from the sum and difference of their tangents. Then ϵ_1 and ϵ_2 could be computed from standard expressions which are quoted, e.g., by Faber and Smith (1968).

This method has the advantage, compared with that of Beattie and Conn (1955), that the detector does not have to be linear. Nor does the sensitivity of the detector and monochromator taken together have to be independent of polarization. Partial polarization of the radiation coming from the source is liable to cause error, but it was verified that this was negligible throughout the visible and ultra-violet. The source used for infra-red work was quite strongly polarized, but fortunately, since the grating monochromator was replaced by interference filters in the infra-red, the detector-plus-monochromator was no longer polarization

sensitive. In the infra-red, therefore, the roles of the polarizer and analyser were reversed.

The measurements were repeated with P alternating between 0 and $-\pi/4$ instead of 0 and $+\pi/4$ and the results averaged; in this way the effects of zero errors in the scales on which P and Q were read were safely eliminated. Some readings were repeated with one or both polarizers rotated through π ; such a rotation did slightly affect the transmitted intensity, probably because the thickness and preferred orientation of the polarizing films were not perfectly uniform, but it did not significantly affect the averaged answers for ψ and Δ . As a further check readings were repeated with P alternating between $\pi/2$ and $\pi/4$, which reverses the sign of the critical settings for Q and the averaged answers were still the same.

The accuracy with which the Q settings could be determined was limited mainly by the inevitable fluctuations of intensity, due particularly to vibration of the specimen surface. It was sufficient to enable ϵ_1 and ϵ_2 , in favourable circumstances, to be determined to within 5% from a single set of measurements. There were, however, other sources of error which could have been systematic. Over some parts of the frequency range the polarizers were not 100% effective. A small correction for imperfect polarization was applied where necessary, and the residual error is unlikely to have exceeded 2% at the worst. Uncertainty in the angle of incidence could be responsible for a further 1 or 2%. In the ultra-violet it is possible that scattered light, which is discussed in the following section, could have introduced yet another error of up to 5%.

§ 3. EXPERIMENTAL DETAILS

3.1. *The Optical System*

The light beam was defined by four horizontal slits, two on each side of the specimen, with dimensions 1 mm $\times$ 20 mm; with a flat reflecting surface only two slits are necessary, but with liquid specimens there is liable to be considerable curvature due to meniscus effects. The slits were separated by 500 mm and therefore defined the angle of incidence to within 0.1° . For work in the visible and ultra-violet it was kept fixed at 70° , which is close to the principal angle of incidence for copper at a wavelength of 0.4μ . In the infra-red it was increased to 80° to improve the accuracy.

Over most of the frequency range the source was a 150 w high-pressure xenon lamp used under d.c. conditions which had a continuous spectrum from 0.2 to 1.4μ , but the readings at 1.01 , 1.26 and 1.604μ were taken using a 150 w tungsten iodide projector lamp. The detector was a photomultiplier in the visible and ultra-violet, a lead sulphide photo-conductive cell in the infra-red. A lens was used in conjunction with the latter to focus the beam onto the sensitive element. The polarizers

were constructed from Polacoat 105 UV for the visible and ultra-violet and from HR Polaroid for the infra-red.

The light beam was chopped at the source with a frequency of 30 sec^{-1} and phase-sensitive detection was used to ensure that light emitted by the molten metal did not affect the readings. The somewhat unconventional placing of the monochromator, after the specimen rather than before, was designed to reduce noise due to this emitted light.

The grating monochromator had a bandwidth of 30 nm. The bandwidth of the interference filters used to select wavelengths in the infra-red was about 0.1μ . Interference filters were used in conjunction with the monochromator at wavelengths between 0.5μ and 0.65μ , where the intensity available in a 30 nm bandwidth was rather low, to reduce the amount of light reaching the detector by scattering from the grating rather than diffraction. Scattered light was also a problem for wavelengths below 0.3μ . No suitable interference filters were available in this region so measurements were made with and without a glass microscope slide inserted in the path of the beam. Since this was effectively opaque below 0.3μ but transparent in the visible region where most of the scattered light was concentrated, the required intensity could be obtained from the difference between the two readings.

3.2. *The Furnace and Vacuum Systems*

The specimens were usually contained in alumina dishes with dimensions $3 \times 4 \times 0.5 \text{ cm}$, though molybdenum dishes were preferred for the experiments on copper and steel ones for silver, since these gave a flatter meniscus. They were surrounded by an alumina tube of length 6.5 cm and diameter 3.5 cm on the outside of which the heating element was wound. This was a coil of molybdenum wire of 1.5 mm diameter which would stand a dissipation of 1600 w, sufficient to raise the temperature of the specimen to about 1600°C . Temperature was measured to within 20°C by a Pt/Pt-Rh thermocouple immediately below the dish. The heater was surrounded by six molybdenum radiation shields, with holes to allow the light beam to pass through. The whole assembly was contained in a stainless-steel can with two side tubes, at the end of which quartz windows were set; the windows had to be about 20 cm from the specimen surface to avoid undue condensation of metal vapour. The side tubes were inclined to the horizontal so that the windows were perpendicular to the beam; it was verified that stresses induced in the windows by evacuating the can did not introduce errors.

Partial evacuation was essential, both to reduce convection losses and to prevent rapid oxidation of the heating element. A diffusion pump enabled the pressure to be reduced to 10^{-5} torr if required, but at this level the rate of evaporation of the specimen and the rate of condensation on the windows were both undesirably high. In practice, therefore, the pressure was maintained at 10^{-1} torr by allowing gas to leak continuously into the can. For nearly all the measurements the gas was pure hydrogen,

designed to reduce oxide films that might have formed on the specimen surface otherwise. Fortunately, the high temperatures at which the measurements were made meant that surface contamination was less of a problem than it is with low melting point metals or with solid ones. Roberts (1960) and Beaglehole (1965) have presented evidence that the oxide film on solid copper decomposes on heating to only 400°C, whether *in vacuo* or in a reducing atmosphere, and it is well known that even if the oxide does not decompose it usually dissolves in the liquid metal when the temperature is high enough. To the eye, signs of surface scum were no longer visible by the time the melting point was reached, for all the metals investigated except aluminium. On aluminium the oxide was more tenacious but it disappeared suddenly on further heating, and at the temperature at which measurements were made (900°C) the surface appeared clean and specularly reflecting.

Two experiments were performed on copper in an atmosphere of argon rather than hydrogen. The results obtained in these runs were systematically lower by about 10%, for both $-\epsilon_1$ and ϵ_2 , than the results of other runs. This is consistent with the presence of a contaminating film (see below), and only the higher values are presented in what follows. Apart from these two cases, no tendency was detected for the results of one run to differ systematically from those of another on the same metal.

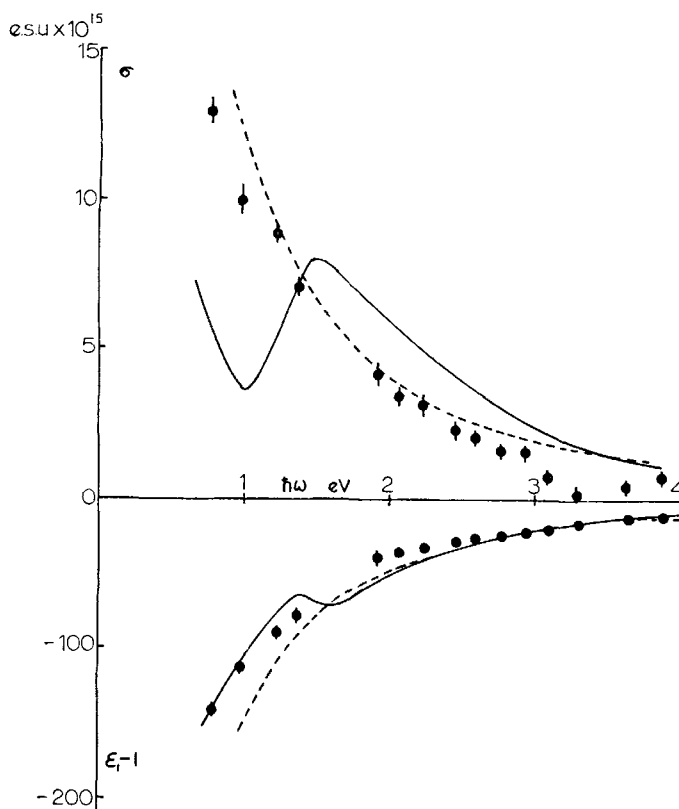
3.3. The Specimens

In optical experiments on liquid metals there is some risk that surface-active impurities may diffuse to the surface and significantly affect the results, even though their concentration in the bulk is quite low. The specimens used in the present work were therefore cut from the purest material available: they were of 4N standard except for Fe (3N), Au (5N) and Cu (3N+ fuse wire or 6N5 Gallard and Schlesinger). More than one specimen was used for each metal, though on some occasions the same specimen was used twice over, being allowed to solidify between runs. For each of the three metals Fe, Ni and Cu two quite different sources were used in preparing the specimens, which therefore presumably had different impurity contents. The reproducibility referred to above is good evidence that the amount of impurity present initially, or introduced into the specimens during use, was in no case sufficient to produce significant effects.

§ 4. RESULTS

The experimental results are shown in figs. 1–7, where $(\epsilon_1 - 1)$ and $\sigma(\omega)$ ($=\omega\epsilon_2/4\pi$) are plotted as functions of photon energy. The temperatures at which the measurements were made are indicated in the figure captions. Values taken from the literature for $\sigma(0)$, the d.c. conductivity of the bulk metal at the same temperature, are also noted. The results of Otter (1961) on liquid Cu and Au, and of Hodgson (1960) on liquid Ag, are included for comparison.

Fig. 1

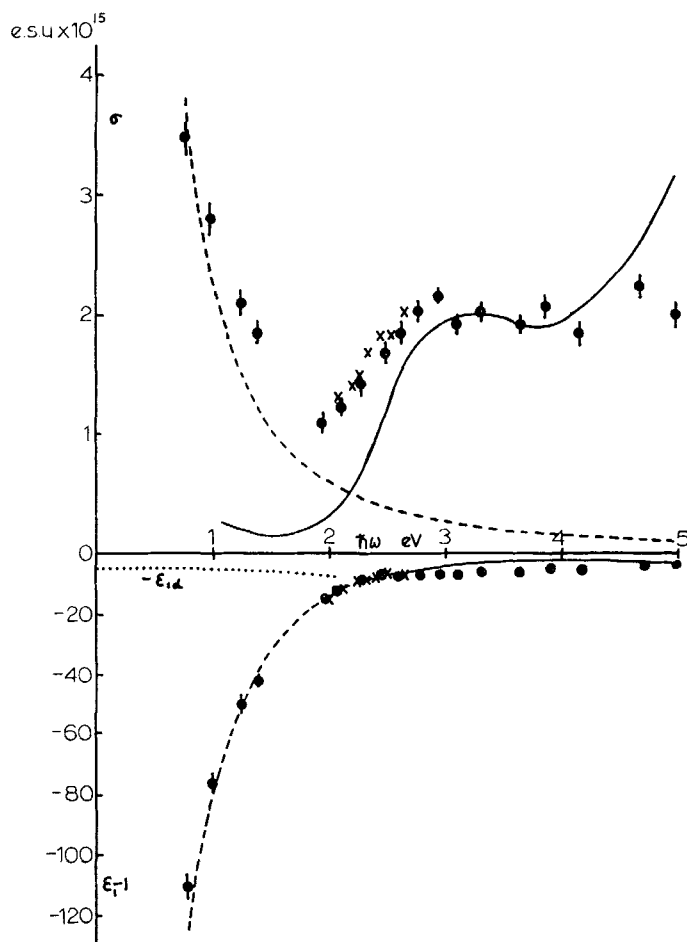


Experimental results for Al: ●, liquid at 900°C; full lines, solid at room temperature (Ehrenreich, Philipp and Segall 1963). The broken curves show the behaviour predicted by the Drude formulae with $n^*/n=1$ and $\sigma(0)=38.2 \times 10^{15}$ e.s.u., the observed d.c. conductivity for liquid Al.

Nearly all the points plotted in these figures represent an average of two or more readings taken in separate runs. The error bars are a rough indication of the standard error for each point, as estimated from the random scatter of the readings. This scatter was no larger than was to be expected from the sources of error mentioned above. Where the error bars are particularly large the circumstances were unfavourable for accurate measurement: the angle of incidence may have been rather far from the principal angle, for example, or the reflected intensity may have been rather low. Only a small amount of pure Au was available so that a small dish was used for this metal; the reflected intensity was therefore cut down by curvature of the meniscus, to such an extent that measurements on Au in the infra-red were not feasible. Supplies of pure Co were also limited and the trouble was aggravated by the fact that the rate of evaporation of Co (and incidentally of Fe and Ni too)

was high. Measurements were not extended into the infra-red for any of the transition metals because it seemed unlikely that the results would justify the efforts required.

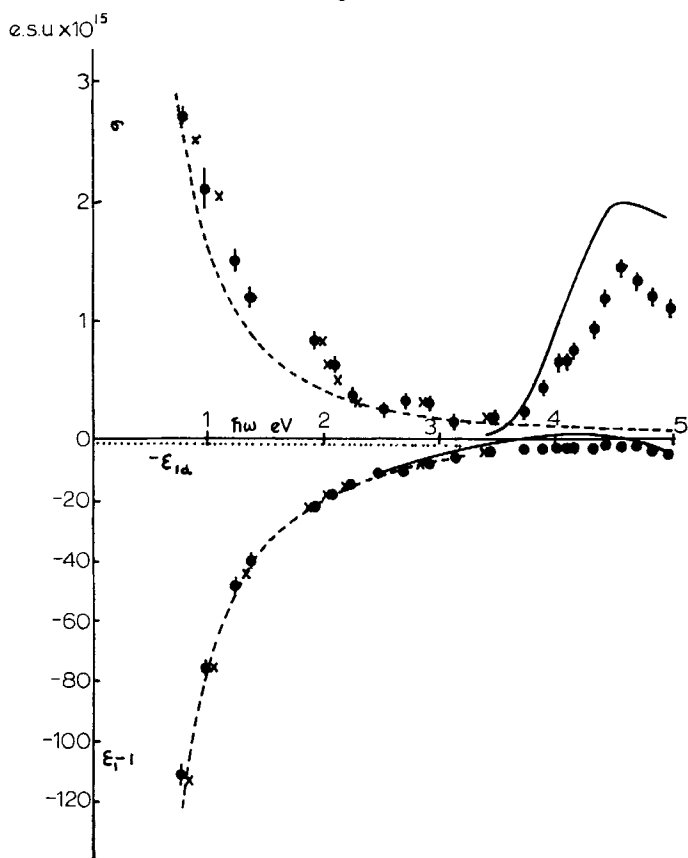
Fig. 2



Experimental results for Cu : ● (present work) and × (Otter 1961), liquid at melting point ; full lines, solid at about 200°C (Beaglehole 1965). The dotted curve shows the behaviour of ϵ_{1d} , estimated by a method described in the text. The broken curves are calculated from the Drude formulac with $n^*/n=0.8$ and $\sigma(0)=42.8 \times 10^{15}$ e.s.u., the observed d.c. conductivity for liquid Cu. The upper broken curve is for σ_c alone : the lower one includes the effects of ϵ_{1d} as well as ϵ_{1c} .

Since no results have previously been reported for solid Co and a comparison between liquid and solid may be fruitful, the Co specimens were allowed to solidify and some measurements carried out just below

Fig. 3



Experimental results for Ag: ● (present work) and × (Hodgson 1960), liquid at melting point; full lines, solid at room temperature (Taft and Philipp 1961, Shulz 1954). The dotted and broken curves have the same significance as in fig. 2, but the latter are calculated using $n^*/n=1.05$ and $\sigma(0)=52.5 \times 10^{15}$ e.s.u., the observed d.c. conductivity for liquid Ag.

the melting point. The results are included in fig. 7. The curves which are plotted to show the solid behaviour of the other metals in figs. 1 to 6 are ones obtained by other workers, mostly at room temperature.

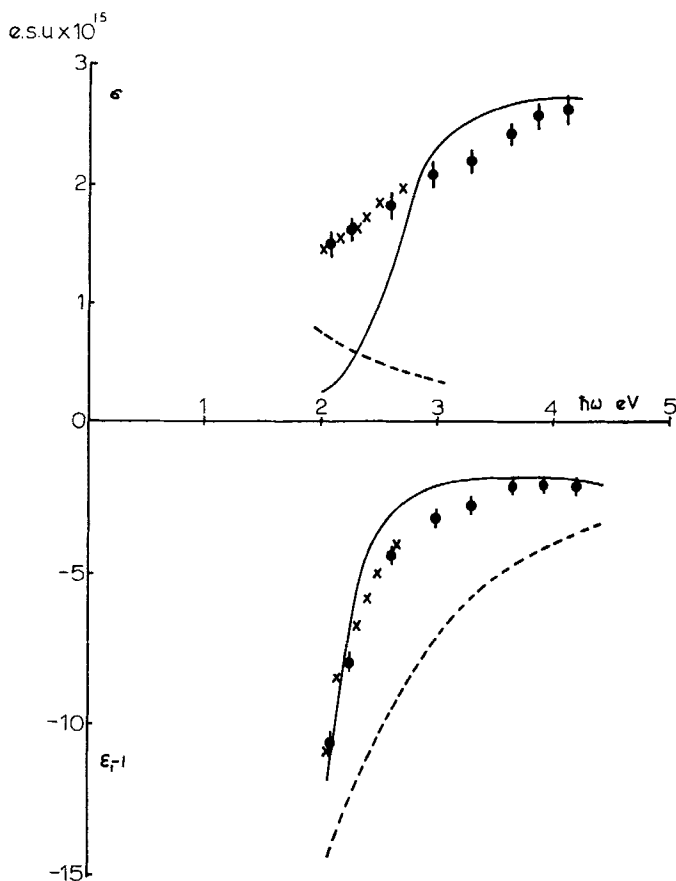
§ 5. DISCUSSION

5.1. Aluminium

For solid Al there is a prominent absorption peak at about 1.5 eV and an associated inflexion in the curve for ϵ_1 ; these two features are linked by the Kramers-Kronig relation:

$$\epsilon_1(\omega) - 1 = 8P \int_0^\infty \frac{\sigma(\omega') d\omega'}{\omega'^2 - \omega^2} \quad . \quad . \quad . \quad . \quad . \quad (2)$$

Fig. 4

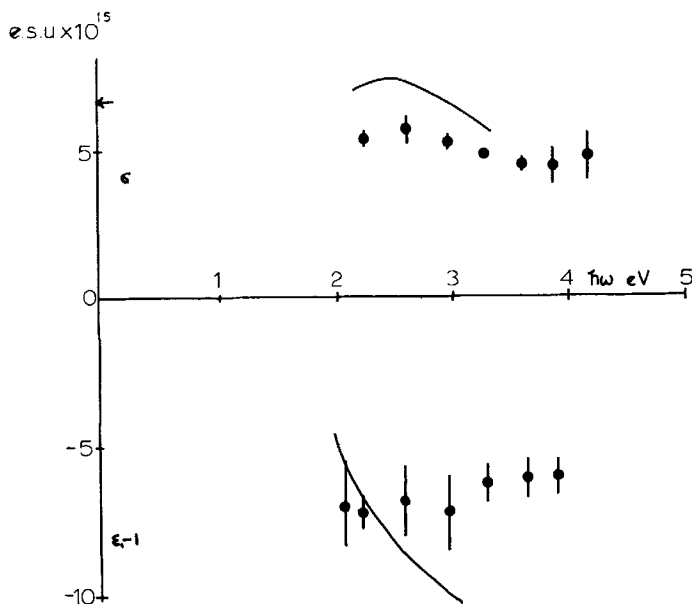


Experimental results for Au : ● (present work) and × (Otter 1961), liquid at melting point ; full lines, solid at about 200°C (Beaglehole 1965). The broken curves are calculated from the Drude formulae with $n^*/n=1.1$ and $\sigma(0)=28.8 \times 10^{15}$ e.s.u., the observed d.c. conductivity for liquid Au. No attempt has been made to estimate ϵ_{1d} , so that the broken curve in the lower half of the diagram depicts $(\epsilon_{1c}-1)$ rather than (ϵ_1-1) and would not be expected to pass through the points. Note the change of scale for this half of the diagram, as compared with figs. 2 and 3.

The first point to notice about the behaviour of the liquid is that it shows no significant structure around 1.5 eV. This is consistent with the general conclusion from recent theoretical work (e.g. Shaw and Smith 1969) that in liquid metals the structure in the density of states curve within a few eV of the Fermi level is always rather small. The x-ray emission spectrum of Al (Catterall and Trotter 1963) was held at one time to indicate that this metal was an exception, with noticeable structure in the density of states for both liquid and solid phases. It

now seems likely, however, that the anomalous shape in the x-ray spectrum of Al has more to do with the way in which the matrix elements for the relevant transition behave than with the density of states.

Fig. 5



Experimental results for Fe : ●, liquid at melting point ; full lines, solid at room temperature (Yolken and Kruger 1965). The arrow marks the d.c. conductivity of liquid Fe.

The broken curves in fig. 1 correspond to the free-electron formulae of (1), with a value of n ($1.55 \times 10^{23} \text{ cm}^{-3}$) corresponding to three electrons per atom and the value of τ ($\hbar/\tau = 0.68 \text{ eV}$) required to reconcile $\sigma(0)$ with the measured d.c. conductivity. These curves come reasonably close to fitting the results, but the discrepancies are clearly significant. How far is it possible to improve the fit if n is replaced by n^* and the latter quantity treated as an adjustable parameter?

It is possible to obtain a value for n^* from ϵ_1 alone, e.g. by plotting $(\epsilon_1 - 1)^{-1}$ against ω^2 and taking the slope of the straight line which should be obtained. This procedure has the advantage that the result should not be distorted by the anomalous skin effect which, according to Holstein (1952) and Dingle (1953), is liable to enhance the apparent value of σ but should not greatly affect ϵ_1 . If, as has sometimes been suggested, τ is a frequency dependent quantity, then in principle the line need not be straight and its mean slope may yield an erroneous value for n^* . We may forget this complication, however, when $(\omega\tau)^2$ is large compared with unity, since ϵ_1 is then independent of τ . For liquid Al it seems that

$(\omega\tau)^2$ only becomes large in the blue and ultra-violet, but nevertheless a plot of $(\epsilon_1 - 1)^{-1}$ against ω^2 does yield an adequate straight line. Its slope corresponds to $n^* = 1.37(\pm 0.1) \times 10^{23} \text{ cm}^{-3} = 0.88(\pm 0.07)n$.

It is difficult to accept this as a meaningful result, on two grounds. Firstly, there is no theoretical sanction for a value of n^* which is less than n . Faber (1966) has considered some corrections to the Drude formulae that should be appropriate for a liquid metal which has no significant band structure but in which, nevertheless, the density of states near the Fermi level departs from the free-electron value by some factor m^*/m . His result for the absorption may be written as† :

$$\sigma(\omega) = \left(\frac{ne^2\tau_z}{m} \right) \left[\frac{(1+\lambda)^2 \left(1 - \frac{1}{24} (\hbar\omega/E_F)^2 + \dots \right)}{1 + \omega^2\tau_z^2(m/m^*)^2(1 - \frac{1}{8}(\hbar/\tau_z E_F)^2 + \dots)} \right], \quad (3)$$

where τ_z is the free-electron relaxation time given by the well-known Ziman formula.

The numerical coefficients $1/24$ and $\frac{1}{8}$ are derived on the assumption that the Fourier components $U(K)$ of the perturbing pseudo-potential in the liquid metal are independent of K , and this is certainly not correct in practice ; but since the Fermi kinetic energy E_F is about 10.5 eV in Al it is clear that the correction terms which involve these coefficients are insignificant over the frequency range of interest here. If (3) is right, therefore, one should be able to fit the Drude formulae to $\sigma(\omega)$, and by implication to $\epsilon_1(\omega)$ also, by choosing

$$n^* = (1 + \lambda)^2 (m^*/m)n \quad (4)$$

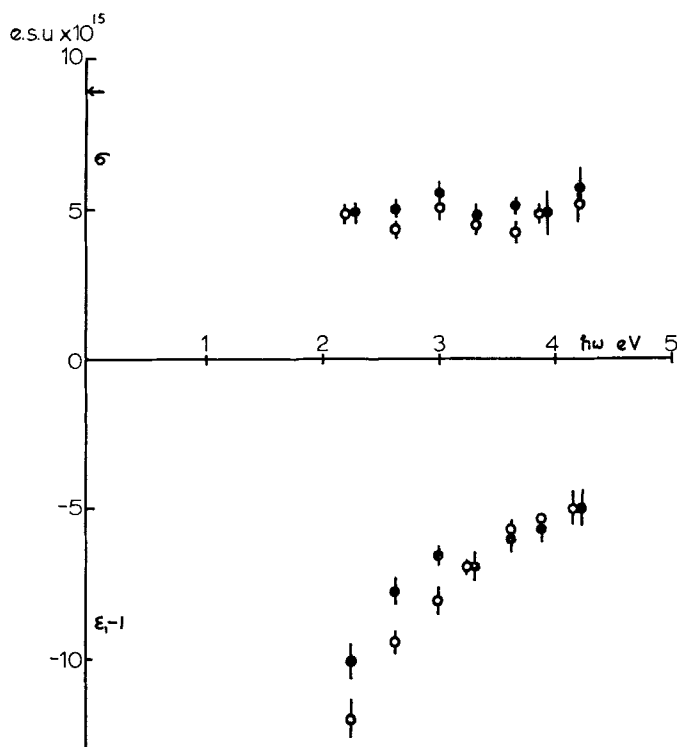
and

$$\tau = \tau_z(m/m^*).$$

The $(1 + \lambda)^2$ correction factor arises from normalization considerations and is likely to be rather greater than unity. Likewise, in so far as (m^*/m) is determined by first-order perturbation of the electrons by the ionic pseudo-potential (including electron correlation and exchange effects), this too is likely to exceed unity for Al : two recent estimates for it are 1.04 (Weaire 1967) and 1.02 (Shaw and Smith 1969). Thus n^* is expected to exceed n .

† I am indebted to Dr. Faber for this formula, which is a simplified and corrected version of what is quoted in his paper. The simplification involves the neglect of a term $\partial A/\partial E$ which recent theoretical work such as that of Shaw and Smith (1969) suggests is trivial. The corrections are required (a) to express the results of a more careful evaluation of terms in $(\hbar\omega/E_F)^2$ and $(\hbar/\tau_z E_F)^2$, (b) to put right a fallacy in Faber's eqns. (91) and (95). The remaining limitations of (3) arise (a) from the neglect of electron-electron interactions in so far as they contribute to (m^*/m) , (b) from the assumption, which is made also in Ziman's theory of $\sigma(0)$, that the Born approximation is valid.

Fig. 6



Experimental results for Co : ●, liquid at melting point ; ○, solid at melting point. The arrow marks the d.c. conductivity of liquid Co.

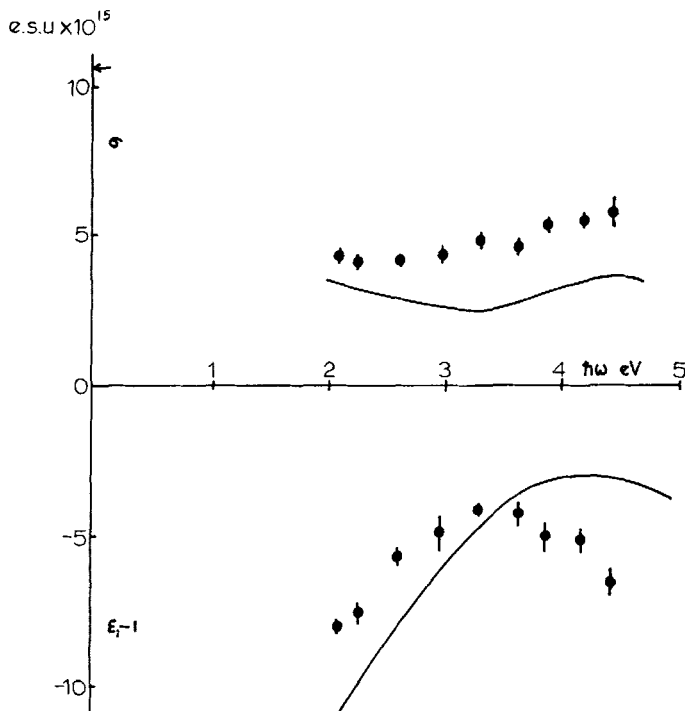
This theoretical conclusion is in agreement with a well-known sum rule, which states that

$$\int_0^\infty \sigma_c(\omega) d\omega = n(\pi e^2/2m). \quad . \quad . \quad . \quad . \quad . \quad (5)$$

Here σ_c is the absorption due to the conduction electrons alone. It has been emphasized by Faber (1966) that for (5) to be valid σ_c must include a negative contribution at high frequencies where, if it were not for the exclusion principle, transitions downwards into core states would be possible. It follows that the area under the absorption curve at relatively low frequencies, where σ_c is due entirely to excitation of the conduction electrons upwards, should appear to extrapolate to a value rather greater than $n(\pi e^2/2m)$, consistent with an enhanced n^* . An apparent value of n^* at low frequencies which is *less* than n can only be reconciled with the sum rule if there is some unsuspected *positive* contribution to σ_c at frequencies beyond the range of measurement. In a liquid metal such as Al the only possible source of extra high-frequency absorption would seem to be electron-electron collisions : these might shorten the lifetime

of excited states well above the Fermi level and enhance $\sigma_o(\omega)$ on that account. Realistic calculations of the importance of such collisions for a liquid metal have not yet been attempted (the results of Gurzhi (1959) seem to be mainly applicable to solids at very low temperatures). It seems unlikely, however, that they could be responsible for a value of n^*/n as low as 0.88 in Al, if in almost all other liquid metals which have been investigated this ratio lies in the range between 1.0 and 1.2.

Fig. 7



Experimental results for Ni : ●, liquid at melting point ; full lines, solid at room temperature (Ehrenreich, Philipp and Olechna 1963). The arrow marks the d.c. conductivity of liquid Ni.

The second ground for rejecting 0.88 as the result for n^*/n is that it does not help to secure a tolerable fit between the Drude formula and the experimental points for σ . A fit can, it is true, be secured between 0.7 and about 2.6 eV by choosing $\hbar/\tau = 0.80 (\pm 0.07)$ eV. But the Drude curve then extrapolates at low frequencies to a value which is only $0.75 (\pm 0.1)$ times the measured d.c. conductivity, while above 2.6 eV it still fails to match the surprising dip which is shown by the experimental points in fig. 1.

One is driven to conclude, therefore, that the data plotted in fig. 1 may not, after all, represent the true optical properties of liquid Al in

the bulk. Does a very thin oxide layer persist on the surface at 900°C despite its apparent cleanliness? It may be shown that the effect of a thin layer of dielectric constant ϵ' and thickness t is expressed by the equation :

$$\epsilon_{\text{app}} = \epsilon \left[1 + i \frac{4\pi t \epsilon' - 1}{\lambda \epsilon'} (\epsilon - \sin^2 \phi)^{1/2} \left\{ 1 - \frac{\epsilon' - \sin^2 \phi}{\epsilon - \sin^2 \phi} \right\} + 0(t^2) \right]. \quad (6)$$

When the angle of incidence is as large as 70° and ϵ_1 is 20 or more, it is a good approximation to replace both $\sin^2 \phi$, and also the factor enclosed by curly brackets in (6), by unity. Then using the Drude expressions for ϵ_1 and ϵ_2 , one may show that

$$\epsilon_{\text{app}} \simeq \epsilon \left[1 + \alpha \left(\frac{\omega\tau}{i + \omega\tau} \right)^{1/2} + 0(t^2) \right], \quad (7)$$

where

$$\alpha = \frac{2t \epsilon' - 1}{c \epsilon'} \left(\frac{4\pi n e^2}{m} \right)^{1/2}; \quad (8)$$

for a 20 Å layer of Al_2O_3 on Al α should be about 0.2†.

Equation (8) means that for $\omega\tau \sim 1$:

$$\epsilon_{1 \text{ app}} \simeq \epsilon_1 (1 - 0.46\alpha), \quad \sigma_{\text{app}} \simeq \sigma (1 - 1.1\alpha),$$

while for $\omega\tau \gg 1$:

$$\epsilon_{1 \text{ app}} \simeq \epsilon_1 (1 - \alpha), \quad \sigma_{\text{app}} \simeq \sigma (1 - 1.5\alpha).$$

These expressions demonstrate that a 20 Å oxide layer could lead to 20% deviations from the Drude curves in the same sense as the deviations revealed by fig. 1. It is clear by inspection of the figure, however, that they do not account in detail for the way in which the observed deviations vary with frequency.

There remains the possibility that the specimen surfaces were clean but that their electrical properties were anomalous, due, e.g. to a regime of anomalously high or low density near the surface. If one is prepared to allow the effective values of n and τ to vary with depth, it naturally becomes possible to explain almost any frequency variation for the apparent optical constants. Unless it is a pure coincidence, however, that $(\epsilon_{1 \text{ app}} - 1)^{-1}$ is as closely proportional to ω^2 as it is, it seems necessary to postulate that n^*/n is almost uniformly equal to 0.88 throughout a surface layer at least 100 Å thick, 100 Å being about the skin depth of liquid Al at the lowest frequencies employed. It is not easy to see why the liquid should remain anomalous to such a depth.

† 20 Å is believed to be the thickness of the oxide layer that forms on freshly evaporated solid Al exposed to air at room temperature (Berning, Hass and Madden 1960). Note that in evaluating (7) one should choose the root of $(\omega\tau/i + \omega\tau)$ which lies in the upper left quadrant of the complex plane.

The detailed interpretation of the data for Al therefore presents considerable difficulties.

5.2. Noble Metals

The absorption edges which are observed for solid Cu, Ag and Au at about 2, 4 and 2 eV respectively are due primarily to excitation of electrons from the full d band to states in the conduction band above the Fermi level. It is also possible in the ultra-violet, however, for conduction electrons to be excited to higher bands, especially across the band gap near the symmetry point L in the Brillouin zone; the observed absorption spectrum results from two mechanisms which are hard to disentangle. The implication of the results plotted in figs. 2-4 is that the d band is rather little affected by melting so that the first mechanism still operates in the liquid phase; the second, however, is presumably suppressed.

If indeed the band structure above the Fermi level disappears on melting, one should expect an increase in the value of n^* required to fit the observations in the infra-red; this is apparent from the discussion about the sum rule in the previous section. A simple argument given by Beaglehole (1966)—see his eqn. (23)—suggests that the clearing of the band gap at L, for example, is liable to increase n^* by a factor of about

$$1 + 1.1(A/4\pi k_F^2), \quad . \quad . \quad . \quad . \quad . \quad . \quad (9)$$

where A is the area over which the Fermi surface for the solid phase makes contact with the [111] faces of the Brillouin zone; this factor could almost be written down intuitively, from the consideration that so long as the gap persists a fraction $(A/4\pi k_F^2)$ of the Fermi surface is sterilized, in the sense that low frequency excitation of the conduction electrons is impossible. From de Haas-van Alphen data one may estimate that the factor described by (9) is about 1.08, 1.04 and 1.07 for Cu, Ag and Au respectively. Since the best values reported for n^*/n in the solid phase are about 0.7, 1.0 and 0.9 respectively† (Beaglehole 1966, Ehrenreich and Philipp 1962) the values to be expected in the liquid are about 0.76, 1.04 and 0.96. These predictions are in satisfactory agreement with the present results, for Cu and Ag at any rate: an adequate fit to the experimental points for $(\epsilon_1 - 1)$ may be achieved (see figs. 2 and 3) by choosing n^*/n to be 0.8 ± 0.05 for Cu‡ and 1.05 ± 0.05 for Ag.

In computing the broken curves for $(\epsilon_1 - 1)$ which are plotted in figs. 2 and 3 account has been taken not only of the contribution due to the conduction electrons, ϵ_{1c} , which the Drude formula sets out to describe, but also of the d-electron contribution, ϵ_{1d} . To estimate $\epsilon_{1d}(\omega)$ it is

† The results for the solid have usually been described in terms of an optical effective mass m_a rather than an effective carrier density, but m/m_a and n^*/n are equivalent.

‡ Note that when the d band overlaps the conduction band the sum rule no longer dictates, as it does for Al, that n^*/n must exceed unity.

necessary to split the observed absorption in the ultra-violet into two parts, σ_c and σ_d , and then to integrate $\sigma_d(\omega')/(\omega'^2 - \omega^2)$ as required by eqn. (2). Since a certain amount of guesswork is needed in effecting the split, and also in extrapolating the resultant curve for $\sigma_d(\omega')$ beyond the range of measurement, the values obtained for ϵ_{1d} are not of high accuracy. They are indicated, for Cu and Ag respectively, by dotted curves in figs. 2 and 3. For Au the data are insufficient to allow ϵ_{1d} to be estimated, even roughly, and hence for this metal it is impossible to determine n^* . It may be noted that in principle—and this applies to Al as much as to the noble metals—one ought to include the contribution to ϵ_1 which is associated with excitation of the inner core electrons, i.e. the so-called core polarizability. But this is small enough to be negligible (Beaglehole 1966).

If n^*/n is 0.8 for Cu and 1.05 for Ag, then for the Drude formula for σ to extrapolate to the measured d.c. conductivity $\hbar/\tau$ must be 0.242 and 0.174 eV respectively, while if n^*/n is say 0.96 for Au $\hbar/\tau$ must be 0.294 eV. These values have been assumed in computing not only ϵ_{1c} (though in the range of measurement $\omega\tau$ is sufficiently large compared with unity for ϵ_{1c} to be insensitive to the choice of τ) but also the curves for σ_c which are plotted in figs. 2, 3 and 4. These curves fall well below the experimental points for Cu and Ag in the infra-red. The extra absorption can scarcely be accounted for by the anomalous skin effect alone. It cannot be dismissed as due to experimental error in the present measurements, because the agreement with the previous work of Hodgson (1960) on liquid Ag is extremely good, and incidentally it has also been noted for the solid metals by Ehrenreich, Philipp and Segall (1963) and by Bennett and Bennett (1966). Presumably it is due either to the effects of electron-electron interaction or to some anomalous property of the surface.

The fact that the Drude curve for σ_c is a poor fit in the infra-red means that it cannot be trusted in the ultra-violet. This makes it difficult to isolate σ_d and to effect a precise comparison between the shape and position of the absorption edges in the liquid and solid phases. There is some indication that the edges are broader in the liquid, but this is probably a trivial effect of the higher temperature of the specimens. The full curves which are plotted in figs. 2 to 4 to show the behaviour of the solid phase were obtained at temperatures some 800°C or more below the melting point. Any feature of the absorption spectrum which is associated at low temperatures with the sharp cut-off of the Fermi distribution at $E = E_F$ is liable to be smeared out over a range of at least $4kT$ when the specimen is heated, and a rise in temperature of 800°C is therefore liable to spread the absorption edges by at least 0.3 eV. As for the position of the edge, it may be shifted to slightly higher frequencies in Ag, but there is no evidence for a shift in the case of Cu and Au.

If the d band in these noble metals is virtually unbroadened by the process of melting and unshifted relative to the Fermi level, then there

is no reason to expect much change in the magnitude of σ_a . In Cu and Au the level of absorption in the ultra-violet is indeed much the same for the liquid as for the solid, at any rate up to about 4 eV. The points for liquid Cu beyond 4 eV fail to show the secondary rise which is displayed by the solid curve, but this is hardly surprising: although the exact explanation for this rise is uncertain (Beaglehole 1966) there is little doubt that it is in some way a consequence of band structure in the conduction electron states above the Fermi surface—it may, for example, be associated with excitation across the band gap near L and therefore be a rise in σ_c rather than σ_a .

It seems likely that the effects which give rise in solid Cu to two distinct peaks in the absorption spectrum occur in solid Ag at much the same energy, so that the two peaks coalesce into one. If this view is correct it is hardly surprising that when Ag melts, and one of the effects is suppressed, the height of the absorption peak is significantly diminished, as is shown to be the case by fig. 3.

5.3. Transition Metals

In Fe, Co and Ni the d band overlaps the Fermi level, the number of unoccupied states within it being equivalent, in the solid state at any rate, to roughly 2.6, 1.6 and 0.6 holes per atom respectively. Calculations (quoted by Ehrenreich, Philipp and Olechna 1963) suggest that in Ni the d band extends some 3 to 4 eV below the Fermi level and this estimate may not be far wrong for Fe and Co too. The situation is complicated below the Curie point by the exchange interaction which splits the d band and is responsible for ferromagnetism. In comparing the results for solid and liquid in figs. 5 to 7 it must not be forgotten that the curves for solid Fe and solid Ni were obtained at room temperature using ferromagnetic specimens. According to Shiga and Pells (1969), however, the absorption spectrum of solid Ni changes only slightly on heating through the Curie point.

The most noticeable feature of the present results is that $\sigma(\omega)$ tends to converge on melting to the same value for all three metals, almost independent of frequency, of about 4.5×10^{15} e.s.u. Considering the complicated nature of the absorption processes that must be involved, both inter-band and intra-band, and the appreciable variation in the d.c. conductivity between the three (from 6.4×10^{15} e.s.u. for Fe to 10.6×10^{15} e.s.u. for Ni) this uniformity of behaviour is a little surprising. It may be remarked that in liquid Cu, which is of course the next metal in the Periodic Table after Fe, Co and Ni, the absorption is also almost independent of frequency, once this is high enough to excite inter-band transitions, but its magnitude is only about 2×10^{15} e.s.u. Thus the absorption in Fe, Co and Ni is roughly twice as great as in Cu. It is tempting to explain this difference by the observation that in the transition metals two types of inter-band transitions are possible instead

of one : electrons can be excited from the sp conduction band into vacant states in the d band, as well as from the d band into vacant states in the conduction band. But until detailed calculations have been performed this simple explanation must be accepted with reserve.

ACKNOWLEDGMENTS

I should like to thank Dr. T. E. Faber for his advice and supervision during the course of this research and the Science Research Council for financial support.

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